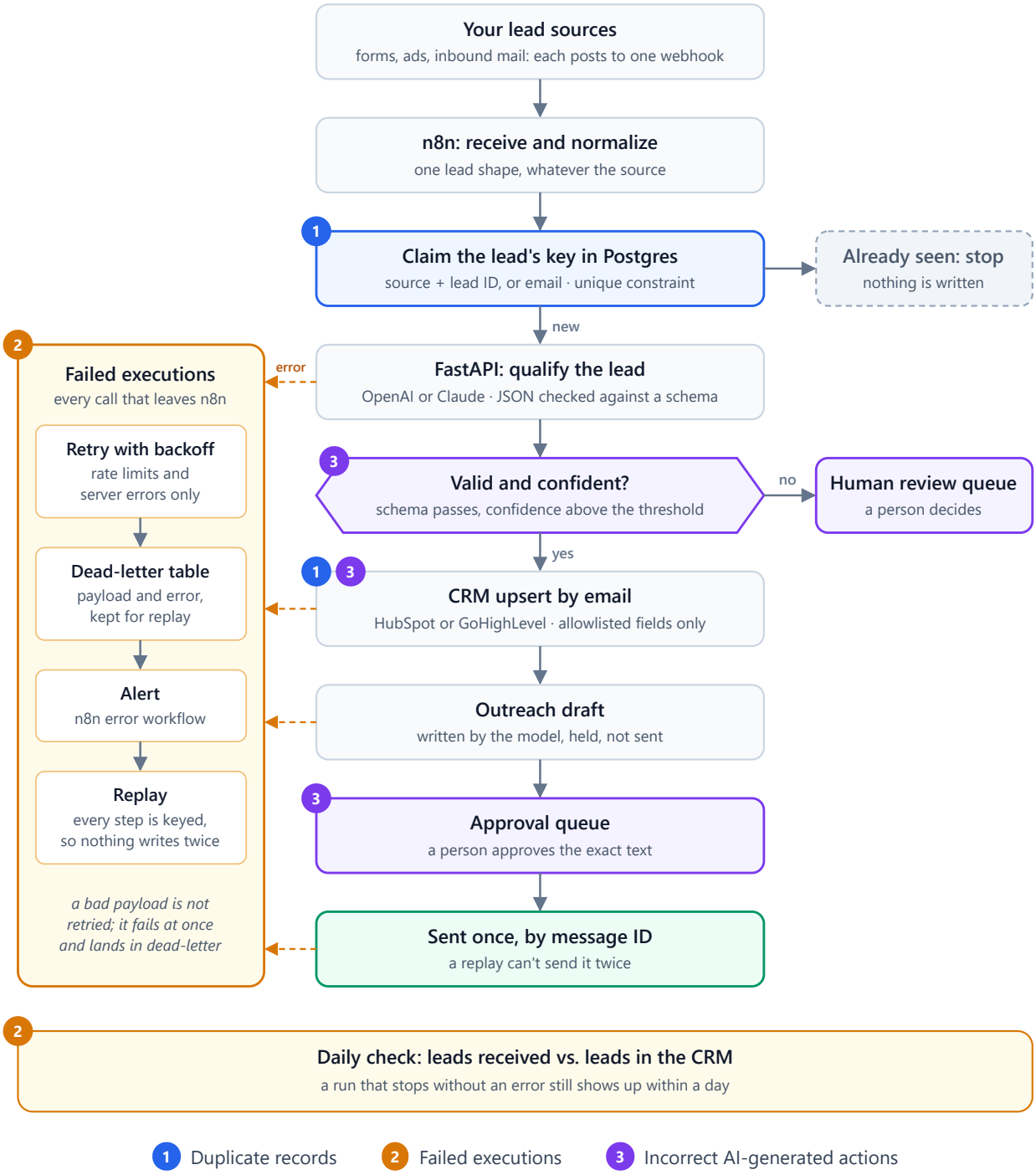
UNPAUSEAI · PROPOSAL BRIEF

Lead pipeline design: where duplicates, failed runs and AI mistakes get stopped

Prepared for the Riyadh team · Matthias, UnpauseAI · 4 November 2026

One path for every lead, with a control at each of the three places you named. n8n moves the lead, FastAPI holds the business logic and the model calls, and Postgres holds the keys, the queues and the dead-letter table.



1 Duplicate records

- Each lead gets one key: the source plus the source's own lead ID, or the normalized email when there is no ID.
- The key goes into Postgres with a unique constraint before anything else runs; a repeated webhook or a replayed run stops there.
- CRM writes are upserts keyed on email. HubSpot's batch upsert takes email as the identifier; GoHighLevel's upsert follows the location's "Allow Duplicate Contact" setting, so that setting is checked before go-live.

2 Failed executions

- Calls retry with backoff on rate limits and server errors only; a bad payload fails at once.
- Whatever still fails is stored in a dead-letter table with its payload and error, and the n8n error workflow sends an alert.
- Replay re-runs a stored item. Because every step is keyed, a replay can't create a second record, task or message.
- A daily check compares leads received with leads in the CRM, so a silent stop shows up within a day.

3 Incorrect AI-generated actions

- The model returns JSON (score, reasons, summary) that is validated against a schema.
- Invalid output, or confidence below the threshold, goes to a person instead of the CRM.
- The AI writes only allowlisted fields (score, summary, draft). Stage, owner and deletions stay with people.
- Every customer message is a draft until someone approves its exact text; an edit afterwards voids the approval, and the message is sent once, by its ID.
- LangGraph comes in where a flow branches on the model's answer; qualification, summary and draft usually fit one validated call.

Delivery

Phase	What ships	Duration	Price
1	Lead intake, duplicate prevention, CRM sync, retries, logging, alerts	about 1.5 weeks	$1,000
2	AI qualification, lead summaries, outreach drafts, approval queue	about 2 weeks	$1,500
3	Follow-ups, tasks, notifications, documentation, Docker deployment	about 1 week	$1,000
	Total, fixed price		**$3,500**

Built before

- **n8n + CRM:** about 20 live n8n workflows against a CRM API for a heating-installation company, with an error handler and a health-digest watchdog.
- **FastAPI + LLM:** a receipt-matching service for a finance team; the model proposes, a person confirms, and nothing guessed reaches the accounting system.
- **Approval gate:** per-message approval with an exact-text check, built and tested for a B2B outreach tool.

Attached: `lead-intake-dedupe-review-workflow.json`, stage one of this design as an n8n workflow: the key claim, the duplicate stop, the review route and the dead-letter route. Sending is left out on purpose; drafts wait for approval.